

# University of Florida

MAD6407

EXAM

AUGUST 18, 2016

Name:

ID #:

Instructor: Maia Martcheva

**Directions:** You have 2 hours to answer the following questions. You must show all your work as neatly and clearly as possible and indicate the final answer clearly. You may not use a calculator.

| Problem | Possible | Points |
|---------|----------|--------|
| 1       | 25       |        |
| 2       | 20       |        |
| 3       | 15       |        |
| 4       | 20       |        |
| 5       | 20       |        |
| Total   | 100      |        |

(1) (25 points) The iteration

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n) - \frac{1}{2}f''(x_n)\frac{f(x_n)}{f'(x_n)}}$$

for solving the equation  $f(x) = 0$  is known as Halley's method.

- (a) Show that the method can, alternatively, be interpreted as applying Newton's method to the equation  $g(x) = 0$ , with  $g(x) = f(x)/\sqrt{f'(x)}$ .

- (b) Assuming  $\alpha$  is a simple root of the equation, and  $x_n \rightarrow \alpha$  as  $n \rightarrow \infty$ , show that convergence is at least cubic.  
Hint: Part (a) might help.

(2) (20 points) Do the following parts which are unrelated.

(a) Derive a Taylor method of order two for the first order initial value problem (IVP)

$$(1) \quad \begin{cases} y' = \frac{y^2}{1+t} \\ y(1) = -\frac{1}{\ln 2} \end{cases}$$

(b) Derive a numerical method for the following second order IVP by replacing the derivatives with centered differences over the mesh  $t_{n+1} = t_n + h$  where  $h = 1/N$ .

$$(2) \quad \begin{cases} y'' + 2y' + y = \cos t & 0 \leq t \leq 1 \\ y(0) = 1 \\ y'(0) = 0 \end{cases}$$

(3) (15 points) Consider a quadrature rule of the form

$$\int_0^1 f(x) dx \approx Af(0) + Bf'(0) + Cf(\gamma) + Df(1).$$

(a) Determine the constants  $A, B, C, D$ , and  $\gamma$  so that the quadrature formula has maximum degree of exactness.

(b) What is the degree of exactness of the formula in part (a)?

- (4) (20 points) For the function  $f(x) = \ln(x)$  for  $x \in [1, 2]$ , find the minimax approximation polynomial of degree one. Give the exact value of the minimax error.

- (5) (20 points) This problem has the following parts.
- (a) Relative to the  $L^2$  norm on the interval  $[1, 3]$ , find the least squares approximation to the function  $f(x) = 1/x$  using polynomials of degree at most one.

- (b) Find the least squares error of the approximation above.